

Network Fundamentals

Introduction to Cisco IOS

Cisco networking devices

L2 Switch

- 2950, 2960



L3 Switch

- 3550, 3560, 3750, 4500, 6500, 7000



Router

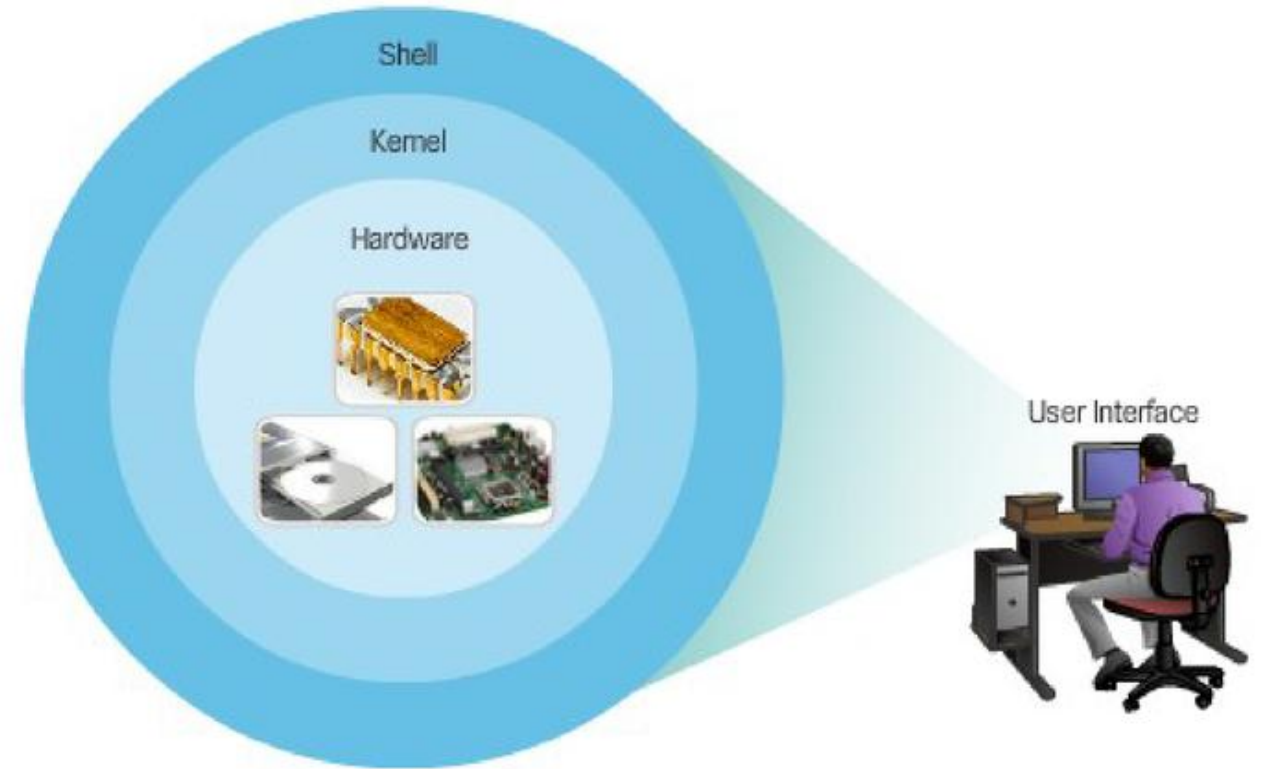
- 2901,



Internetwork Operating system (ios)

Shell: Interaction with user

Kernel: Interaction with hardware

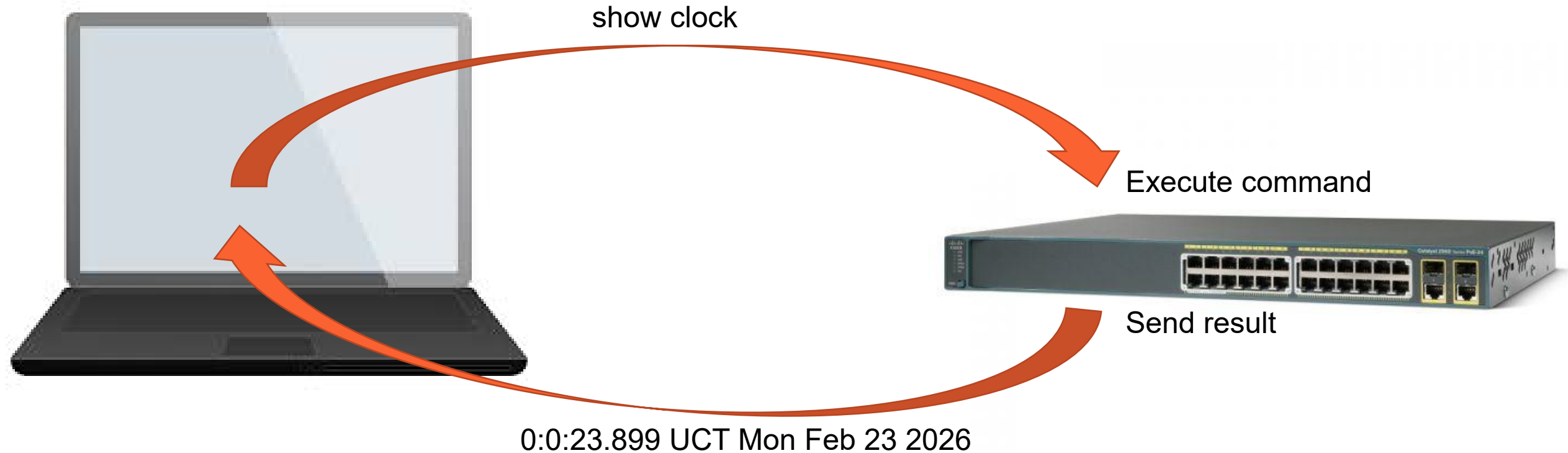


Interacting with the ios

- No monitor or keyboard/mouse on network devices
- Send commands to network devices using a terminal program on a computer
- Command is sent over **serial, network or AUX connection** to the network device and executed on network device



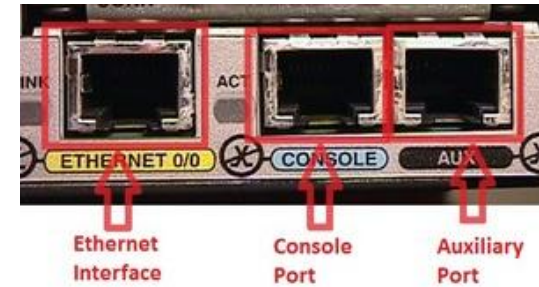
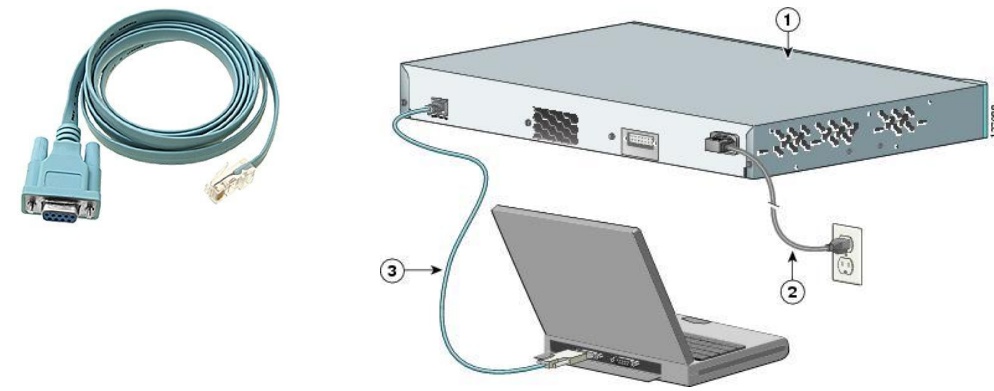
Interacting with the ios



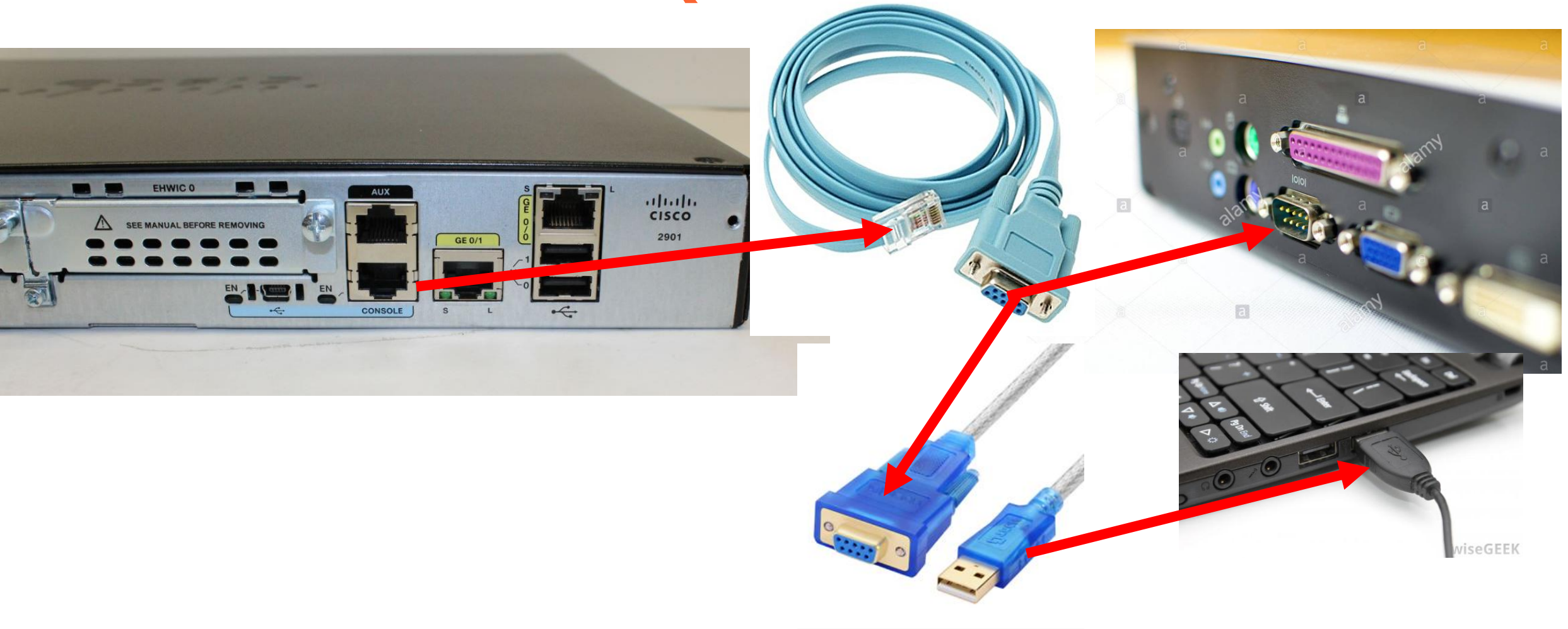
- Commands and results send over serial, network or AUX connection between computer terminal and device

Access IOS CLI

- Console: **direct** physical connection
- AUX: phone line connection
- Ethernet, Secure shell (ssh):
secure **network** connection
- Ethernet, telnet: unsecure
network connection

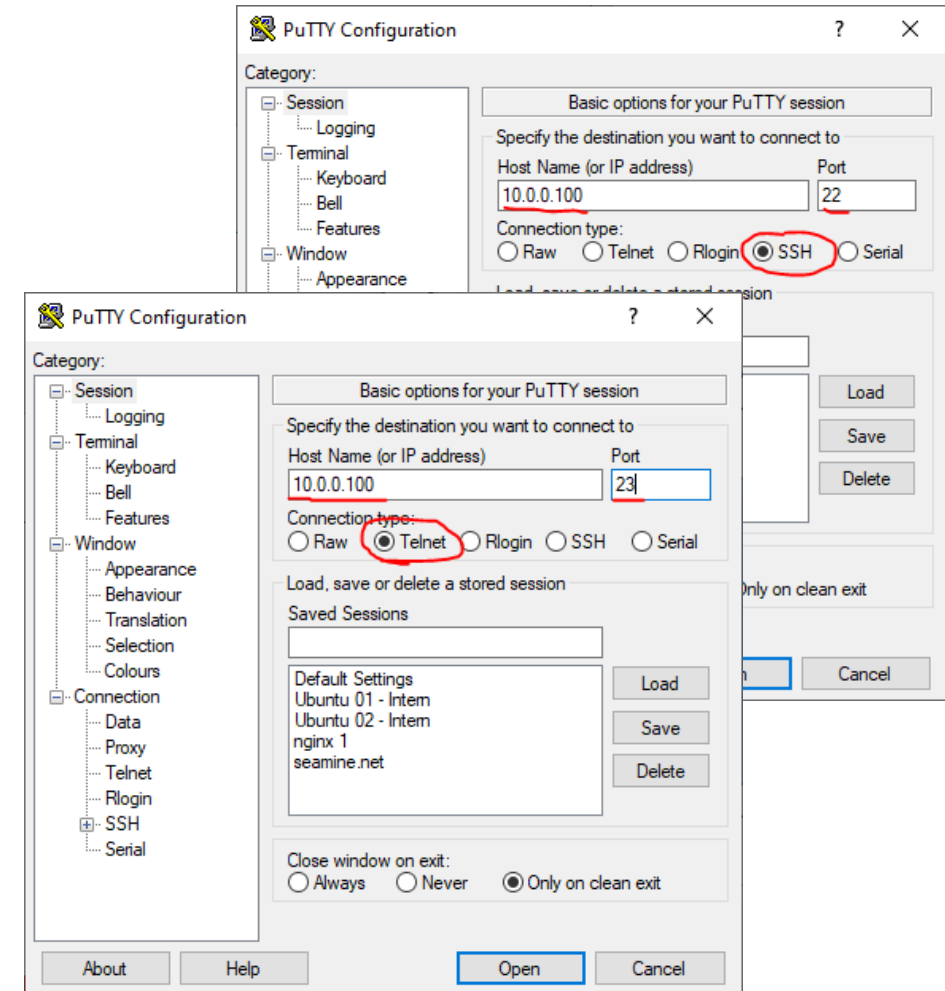
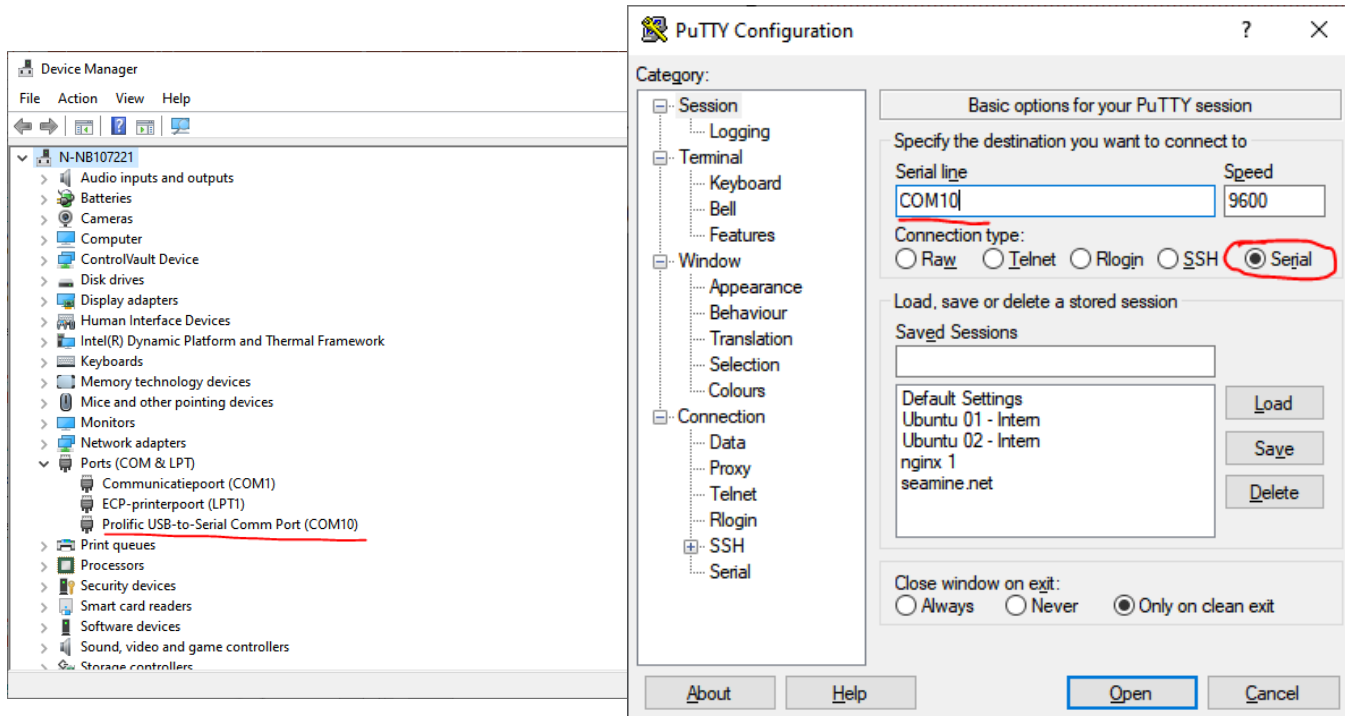


Console access (serial)



Terminal programs Windows

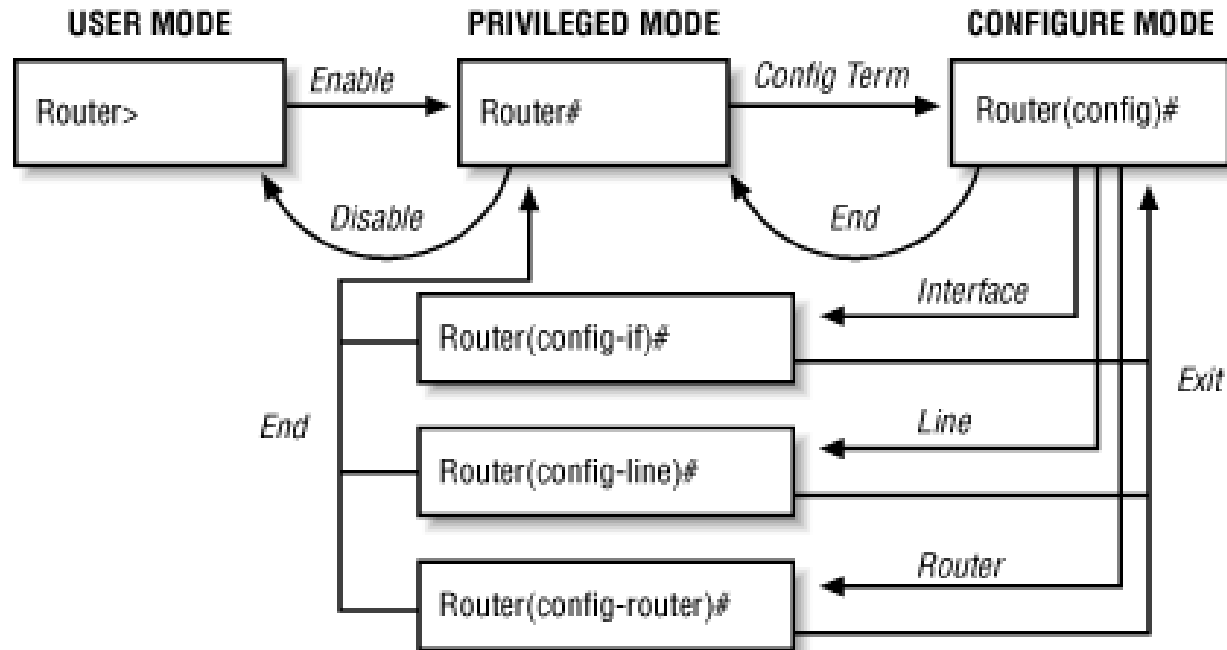
Putty, Tera Term, secureCRT



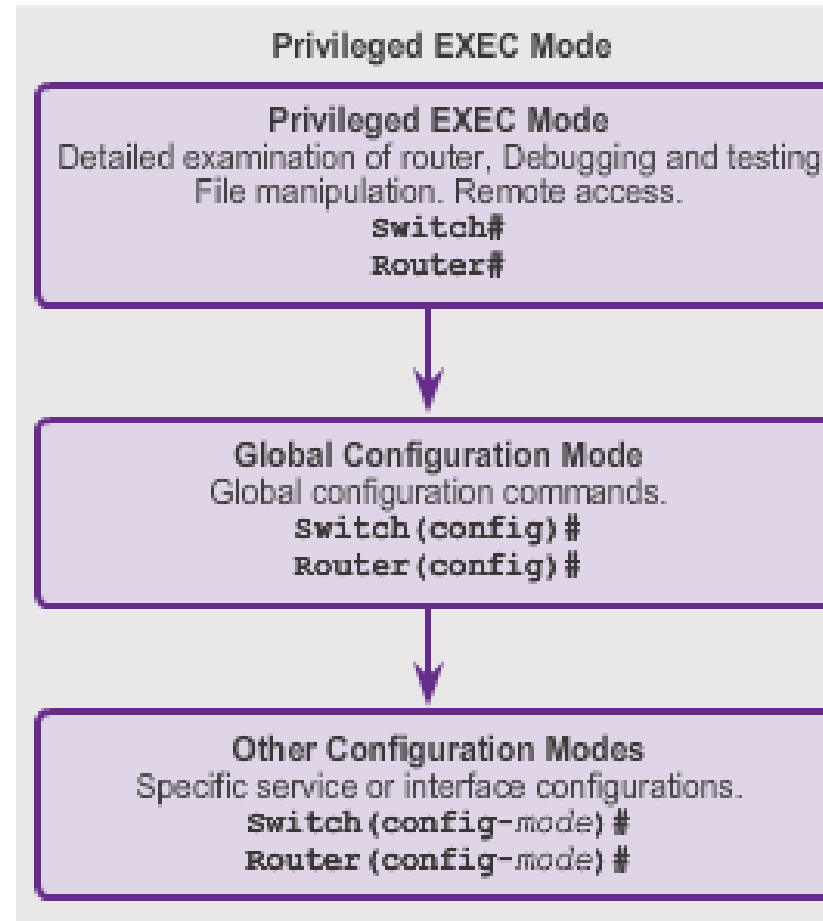
Terminal programs Linux/MAC

- Find correct device
 - `ls -l /dev | grep "ttyUSB"`
- Connect to serial using screen
 - `screen /dev/ttyUSB0 9600`
- Detach from screen session
 - `Ctrl + a, Ctrl + d`
- Reattach to screen
 - `screen -r`

Cisco IOS command modes



Global Configuration Mode and Submodes



Global Configuration Mode and Submodes

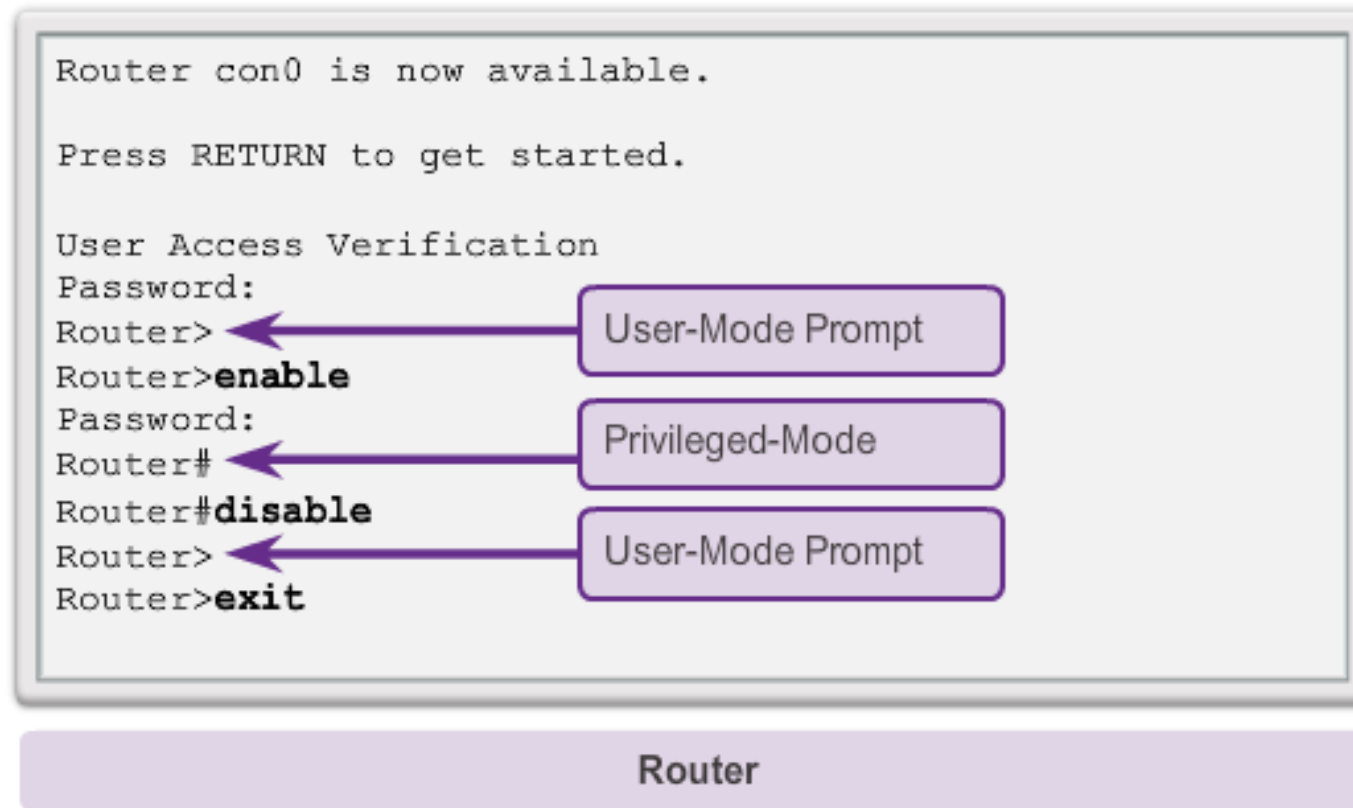
IOS Prompt Structure

```
Router>ping 192.168.10.5  
  
Router#show running-config  
  
Router(config)#Interface FastEthernet 0/0  
  
Router(config-if)#ip address 192.168.10.1 255.255.255.0
```

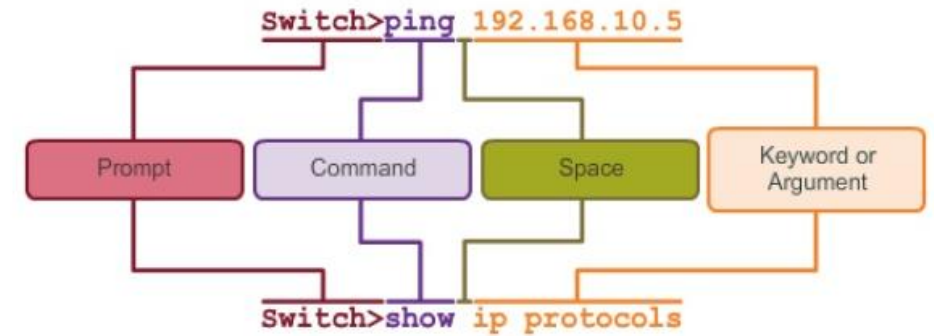
The prompt changes to denote the current CLI mode.

```
Switch>ping 192.168.10.9  
  
Switch#show running-config  
  
Switch(config)#Interface FastEthernet 0/1  
  
Switch(config-if)#Description connection to WEST LAN4
```

Navigating between IOS Modes



Command structure



Use ? to get help

- Switch>? Get all commands
- Switch>p? Get all commands starting with p
- Switch>ping ? Get help about next part of command

Hotkeys

- Tab Complete command
- Arrow up Get previous command

Context Sensitive Help

Context Sensitive Help

```
Switch#cl?  
clear clock  
  
Switch#clock set ?  
hh:mm:ss Current Time  
  
Switch#clock set 19:50:00 ?  
<1-31> Day of the month  
MONTH Month of the year  
  
Switch#clock set 19:50:00 25 June 2012  
Switch#
```

Command options - display a list of commands or keywords that start with the characters **cl**

Command explanation - the IOS displays what command arguments or variables can be next, and provides an explanation of each

Command explanation with more than one argument or variable option

Command Syntax Check

```
Switch#>clock set  
% Incomplete command.  
Switch#clock set 19:50:00  
% Incomplete command.
```

The IOS returns a help message indicating that required keywords or arguments were left off the end of the command.

```
Switch#c  
% Ambiguous command: 'c'
```

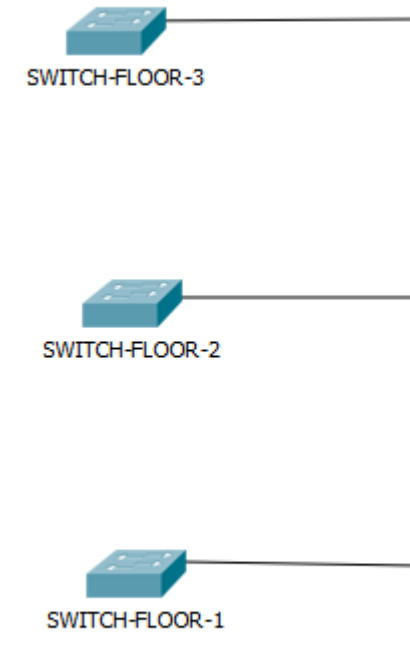
The IOS returns a help message to indicate that there were not enough characters entered for the command interpreter to recognize the command.

```
Switch#clock set 19:50:00 25 6  
^  
% Invalid input detected at '^'  
marker.
```

The IOS returns a "^" to indicate where the command interpreter can not decipher the command.

Device names

- Switch# **configure terminal**
- Switch(config)# **hostname Sw-Floor-1**
- Sw-Floor-1(config)#



Securing Device Access

The passwords introduced here are:

- **Enable password** - Limits access to the privileged EXEC mode
- **Enable secret** - Encrypted, limits access to the privileged EXEC mode
- **Console password** - Limits device access using the console connection

Note: During our labs, use **cisco** or **student** as your passwords!

passwords

Set password for console connection:

- Sw-Floor-1# **configure terminal**
- Sw-Floor-1(config)# **line console 0**
- Sw-Floor-1(config-line)# **password YourPassword**
- Sw-Floor-1(config-line)# **login**
- Sw-Floor-1(config-line)# **end**

Set password to navigate to privileged mode

- Sw-Floor-1# **configure terminal**
- Sw-Floor-1(config)# **enable secret YourPassword**
- Sw-Floor-1(config)# **exit**

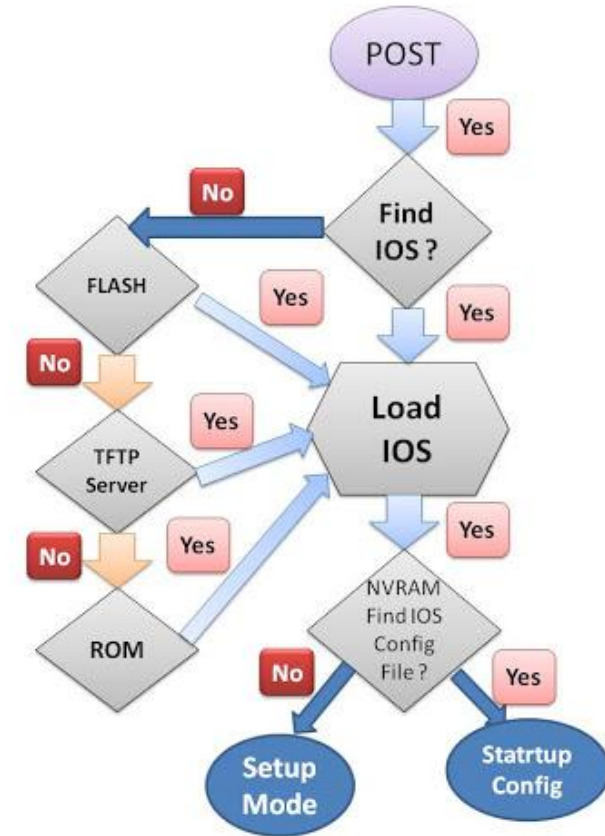
- Encrypt password
 - Sw-Floor-1# **configure terminal**
 - Sw-Floor-1(config)# **service password-encryption**

Configuration files

- **Startup-config:** This is the saved configuration file that is stored in NVRAM. It contains all the commands that will be used by the device upon startup or reboot.
- **Running-config:** This is stored in Random Access Memory (RAM). It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately.
- Switch# **show running-config**

Memory and boot process

Memory	Volatile / Non-Volatile	Stores
RAM	Volatile	- Running IOS - Running configuration file - IP routing and ARP tables - Packet buffer
ROM	Non-Volatile	- Bootup instructions - Basic diagnostic software - Limited IOS
NVRAM	Non-Volatile	Startup configuration file
Flash	Non-Volatile	- IOS - Other system files



Config files

- Changes in configuration are made in the running-config file.
- **S1#reload** command will restart the device. Running config is lost, IOS loads startup config
- **S1#copy running-config startup-config** command will save the running config to the startup config
- **S1#write erase** command will remove the startup config

Show commands

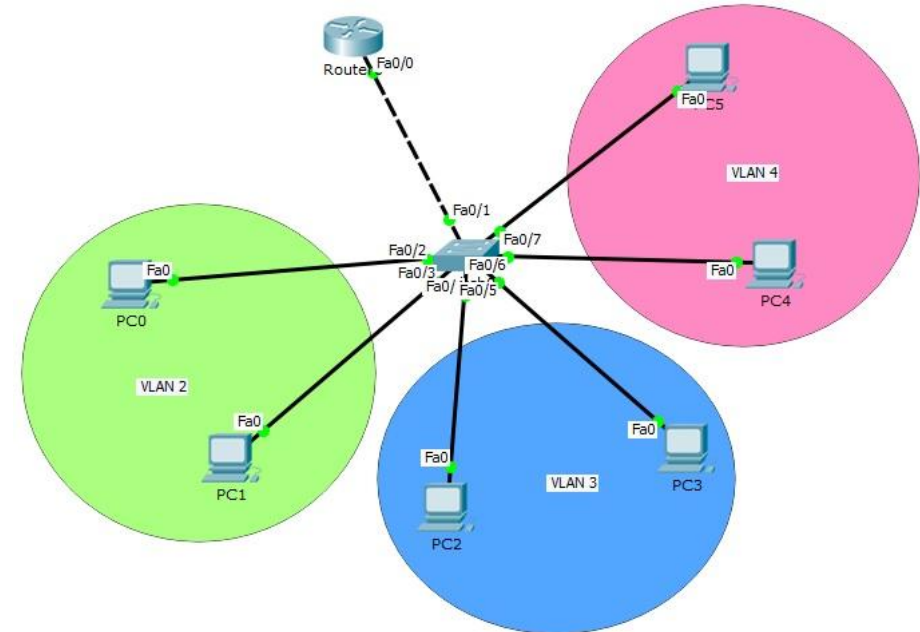
- S1#show running-config
- S1#show startup-config

- S1#show interface
- S1#show interface gi0/1
- S1#show ip interface brief

- S1#show vlan

VLAN's

- The physical ports on a switch can be virtually spit into multiple Virtual LAN's (VLANs)
- Each physical network port can be assigned to access a VLAN



Switch config ip address: VLAN 1

- By default, all ports are member of vlan 1
- Virtual interface
- You can set an ip address on a virtual interface, not on a port
 - S1(config)#interface vlan 1
 - S1(config-if)#ip address 10.0.0.1 255.255.255.0
 - S1(config-if)#no shutdown
- Virtual ip address is accessible by all devices connected to ports that are a member of that virtual interface
 - S1#show vlan

Router config ip address

By default, all router ports are shut down

- R1#show ip interface brief → lists all the available network int
- R1#configure terminal
- R1(config)#interface gi0/0
- R1(config-if)#ip address 10.0.0.1 255.255.255.0
- R1(config-if)#no shutdown

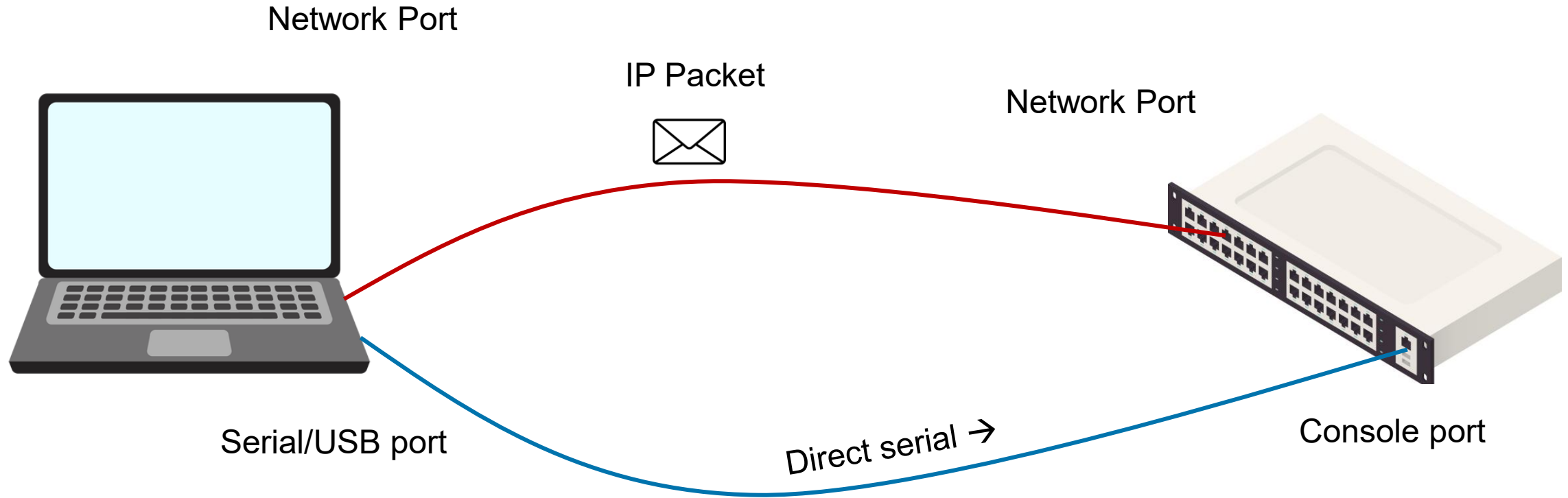
Serial commands

- Serial communication over direct serial connection
- Single connection
No network needed
- Connected to serial ports

telnet/ssh commands

- Communication is packed in an IP packet (source/destination ip and mac)
- Network connection
- Multiple users possible at the same time (vty 0 15)
- Network needs to be configured, devices need to be reachable over the network
- Connected to network ports

Serial vs Telnet/ssh

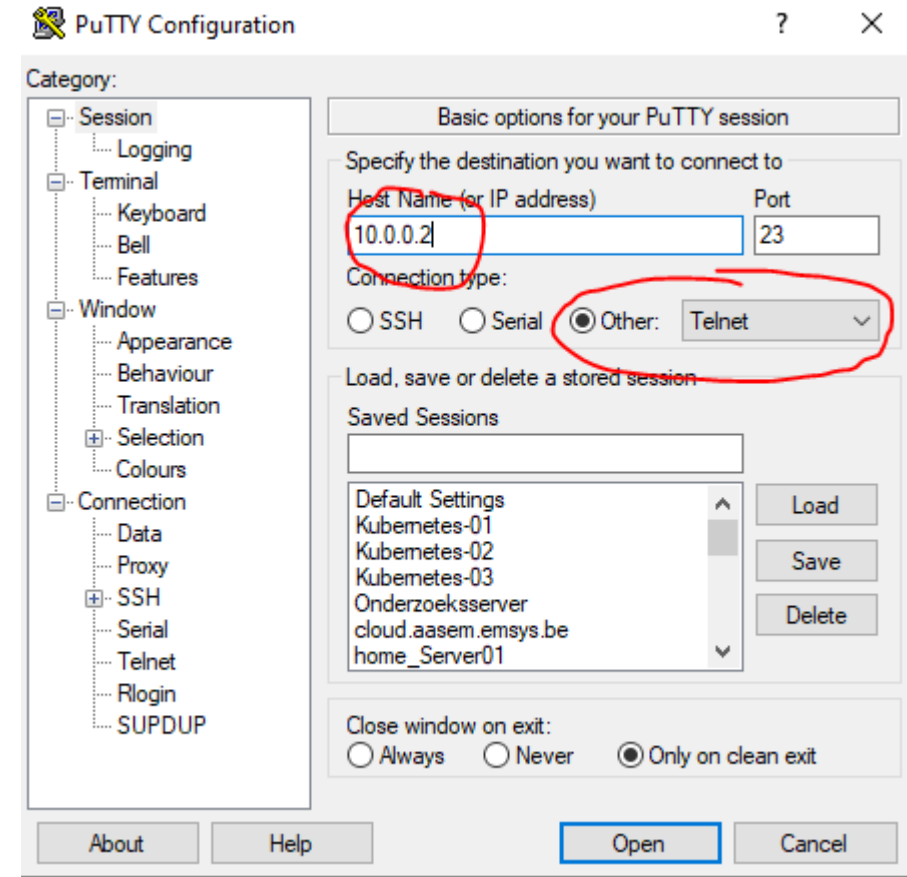


Telnet

- Send commands to device over network
- vty 0-15: 16 available network connections

Vty (telnet and ssh)

- Sw-Floor-1# **configure terminal**
- Sw-Floor-1(config)# **line vty 0 15** <- configure all 16
- Sw-Floor-1(config-line)# **password <pw>**
- Sw-Floor-1(config-line)# **login**
- Sw-Floor-1(config-line)# **end**



ssh

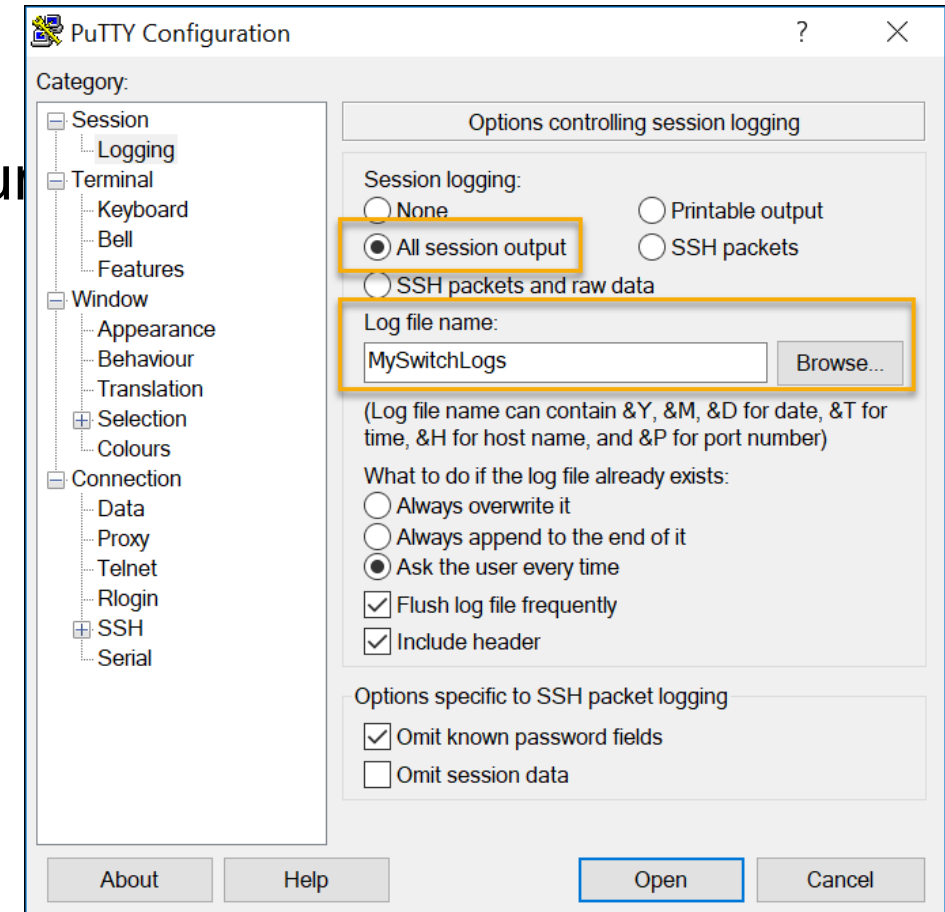
- hostname carter ← Set hostname on switch
- aaa new-model ← set username and password
- username cisco password 0 cisco
- ip domain-name rtp.cisco.com ← Set domain
- crypto key generate rsa
- ip ssh time-out 60
- ip ssh authentication-retries 2
- line vty 0 4 ← Select vty lines
- transport input ssh ← enable ssh on line

Backup via console

Enable Session logging in Putty

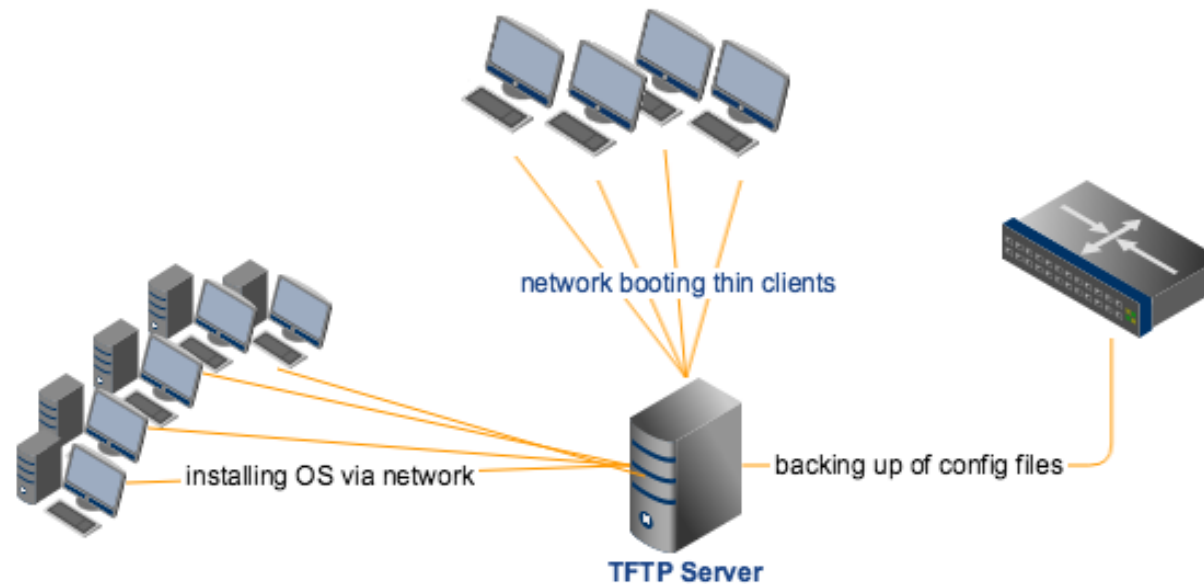
S1#show run command in the cli will output the current configuration

Disable Session logging in Putty



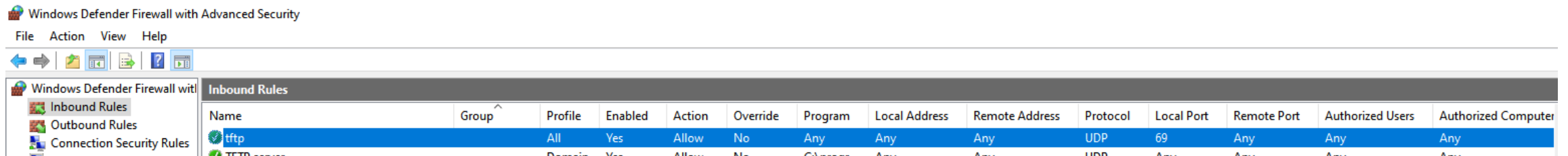
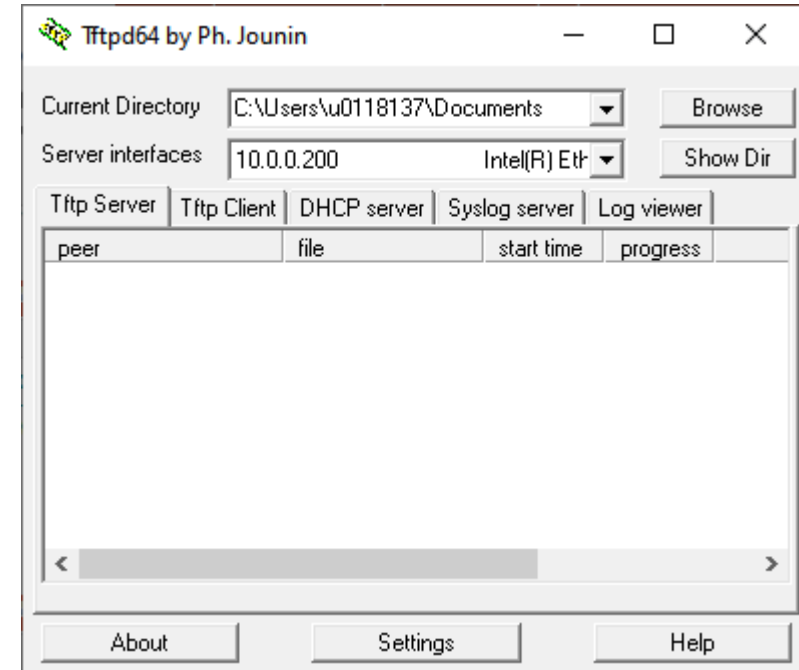
Trivial file Tftp

- Simple, lightweight file transfer protocol
- Client – server communication



Backup via tftp server

- Tftpd64 is tftp (trivial file transfer protocol) server software
- Change 'current directory' to writable folder
 - S1#copy run tftp
- Allow port 69 on your firewall



Hot Keys and Shortcuts

Tab - Completes the remainder of a partially typed command or keyword

Ctrl-R - Redisplays a line

Ctrl-A - Moves cursor to the beginning of the line

Ctrl-Z - Exits configuration mode and returns to user EXEC

Down Arrow - Allows the user to scroll forward through former commands

Up Arrow - Allows the user to scroll backward through former commands

Ctrl-Shift-6 - Allows the user to interrupt an IOS process such as **ping** or **traceroute**.

Ctrl-C - Aborts the current command and exits the configuration mode